

Ajna XR VR Headset

For installing any app on the headset using an apk file:

1. Download the .apk file for the app on your laptop, connect the headset to your laptop with a cable and then transfer the file to any of the folders (for eg. Downloads).
2. On the headset open Resource centre located in the bottom bar. Click on the Path button located on the top.
3. Go to the Downloads folder and click on the .apk file you transferred.
4. Click OK when it asks to install the apk.

Using the Aurora Store:

1. First install Aurora Store on the headset using the above method. The apk file can be found here <https://auroraoss.com/files> under AuroraStore/Latest/latest.apk.
2. Open the Aurora Store on the headset and install any app you want.

Streaming the drone's video feed on the headset:

1. Install VLC Media Player on the headset using either of the options above.
2. Open New Stream under the More option.
3. Enter any rstp link you want and stream.

Suparna 5G Drone

The drone works well with iDronam but since it's a paid software and not everyone can install it on their laptops we thought it's better to use an open source alternative. We went with QGroundControl since it's available on all platforms (Windows, Linux, macOS, Android). You can download it from here

https://docs.qgroundcontrol.com/master/en/qgc-user-guide/getting_started/download_and_install.html.

Fetching the drone's feed

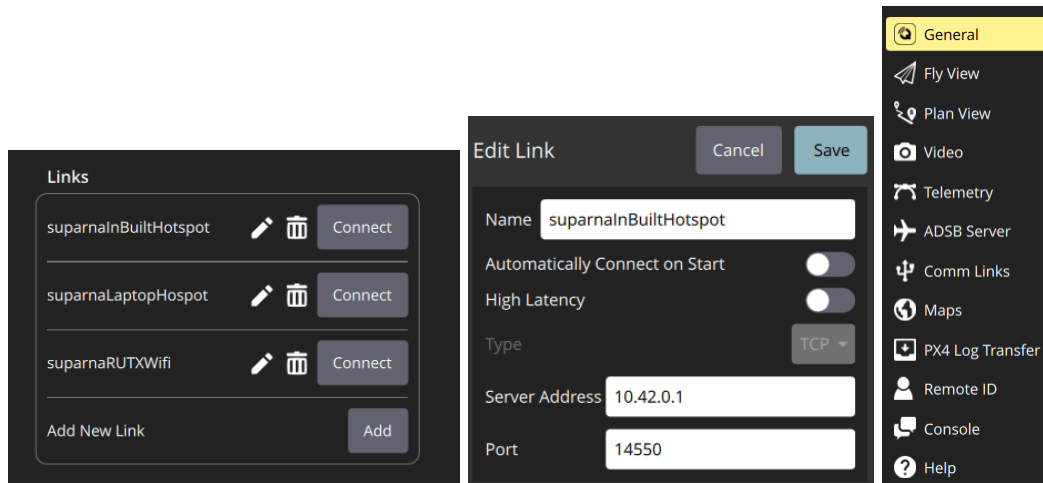
1. The drone broadcasts the stream at rtsp://<drone-ip>:10000/drone_cam (for example: rtsp://[10.42.0.1:10000/drone_cam](#) - when in Hotspot Mode).
2. Stream Trigger: No specific trigger or MAVLink heartbeat is required. The stream initializes and begins transmitting automatically once the drone is powered on.
3. Network Mode: As long as you can connect to the drone's IP, the RTSP stream will be accessible.

Connecting to drone using QGroundControl

The drone connects only using a TCP connection. To add a link to the drone:

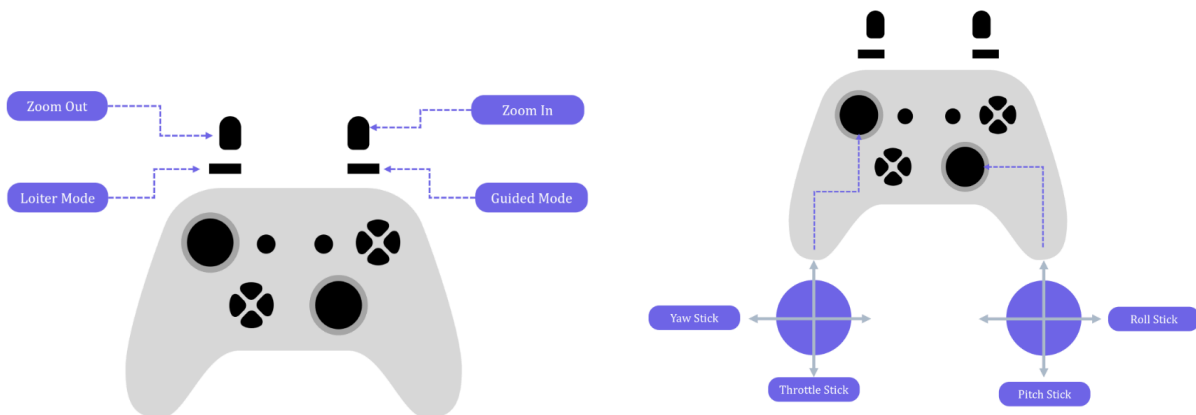
1. Go to the QGroundControl Application Settings. Go to the Comm Links section.
2. Click on Add New Link.
3. Select the type of connection as TCP.

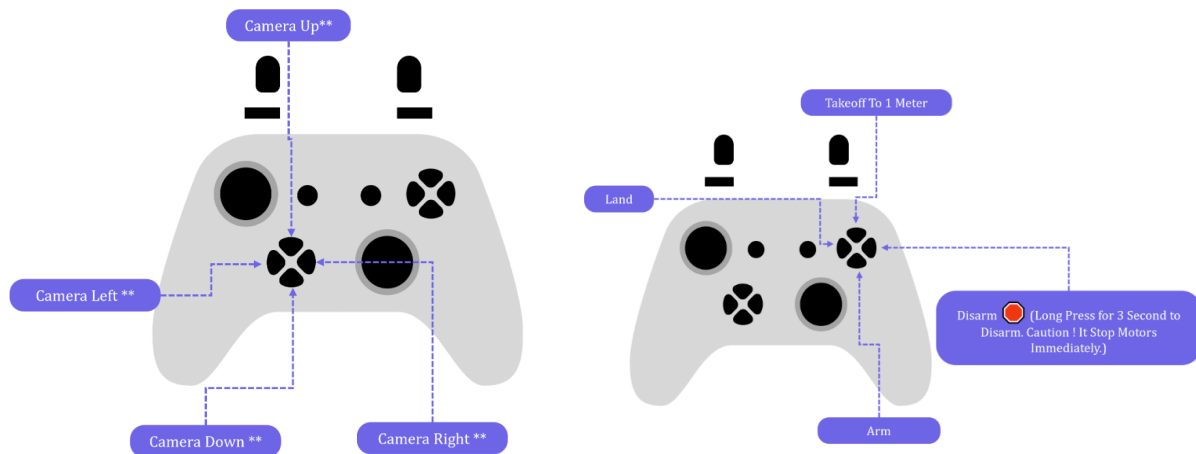
4. Enter port as 14550.
5. In the Server Address type the IP Address of the drone (for example 10.42.0.1 in case you are connecting to the drone using the drone's hotspot itself).
6. Save and Connect.



After connecting the drone you'll need to:

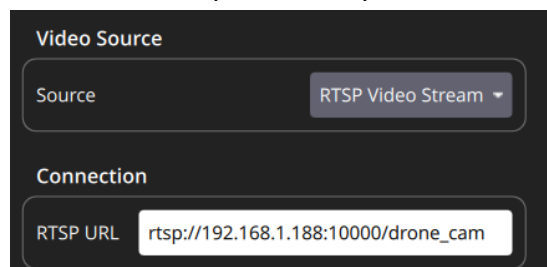
1. Calibrate your joysticks which can be found in Vehicle Configuration → Radio.
2. Assign some function to the buttons of your controller.
3. Disable Hardware Safety Switch inside Vehicle Configuration → Parameters. The drone doesn't actually have a physical hardware safety switch so we need to disable this option.





Streaming drone's video on QGroundControl

1. Go to Application Settings. Go to the Video section.
2. Select source as RTSP Video Stream.
3. Enter `rtsp://<drone-ip>:10000/drone_cam` in the RTSP URL section.

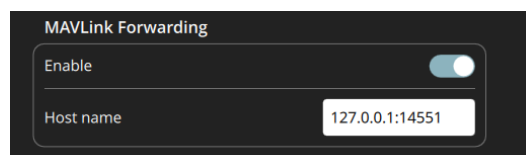


MAVLink Forwarding

This allows you to receive useful data related to the drone such as battery level, flight mode and if the drone is armed or not. To enable this:

1. Go to Application Settings in QGroundControl. Go to the Telemetry section.
2. Enable MAVLink Forwarding.
3. Enter `127.0.0.1:14551` as the host name.

When you use `127.0.0.1`, you are telling the computer not to send the data out over Wi-Fi or the internet, but to loop it right back to itself. An IP address gets the data to the right computer, but a computer has thousands of "doors" or "slots" called Ports. 14551 is the specific slot QGroundControl will release the MAVLink data into.



Ffmpeg installation

Download ffmpeg-release-essentials from <https://www.gyan.dev/ffmpeg/builds/>. Extract the zip file and add the bin folder found inside the extracted folder to your computer's path so that you can run ffmpeg as a command in the command prompt.



release builds			
latest release	version: 8.1	2026-03-16	
ffmpeg-release-essentials.7z	32 MB	.ver	.sha256
ffmpeg-release-essentials.zip	103 MB	.ver	.sha256
ffmpeg-release-full.7z		.ver	.sha256
ffmpeg-release-full-shared.7z		.ver	.sha256

Run the command below to overlay telemetries found inside hud_data.txt onto the drone's video feed and transmit it again on rtsp://localhost:8554/fpv_hud. To stream this video on any other device replace localhost with the computer's IP address on which you are running this command. Run this command after running mediamtx.

```
ffmpeg -rtsp_transport tcp -i rtsp://drone_ip:10000/drone_cam -vf "drawtext=textfile=hud_data.txt:reload=1:fontcolor=white:fontsize=36:box=1:boxcolor=black@0.6:x=20:y=20" -c:v libx264 -preset ultrafast -tune zerolatency -f rtsp rtsp://localhost:8554/fpv_hud
```

Mediamtx installation

Download mediamtx from <https://github.com/bluenvron/mediamtx/releases>. Extract the zip file and add the extracted folder to Path so that you can run mediamtx as a command in the command prompt.

Initially you might get this warning about mediamtx not being able to find a configuration file. Just copy and paste the configuration file that came with the download (for e.g. mediamtx.yml) in the search location. Next time edit this configuration file for configuring mediamtx which we'll do next.

```
PS C:\Users\ARYAN> mediamtx
2026/04/27 11:03:28 INF MediaMTX v1.17.1, windows, amd64
2026/04/27 11:03:28 WAR configuration file not found (looked in C:\Users\ARYAN\rtsp-simple-server.yml, C:\Users\ARYAN\mediamtx.yml), using an empty configuration
```

Since we are transmitting the video on rtsp://localhost:8554/fpv_hud we need to add fpv_hud as a path in mediamtx.yml. You can find paths at the end of the configuration file. Just add "fpv_hud:" and nothing else. Save and exit and run mediamtx in the command prompt to start your server.

```
# Path settings

# Settings in "paths" are applied to specific paths, and the map key
# is the name of the path.
# Any setting in "pathDefaults" can be overridden here.
# It's possible to use regular expressions by using a tilde as prefix,
# for example "~^(test1|test2)$" will match both "test1" and "test2",
# for example "~^prefix" will match all paths that start with "prefix".
paths:
  fpv_hud:

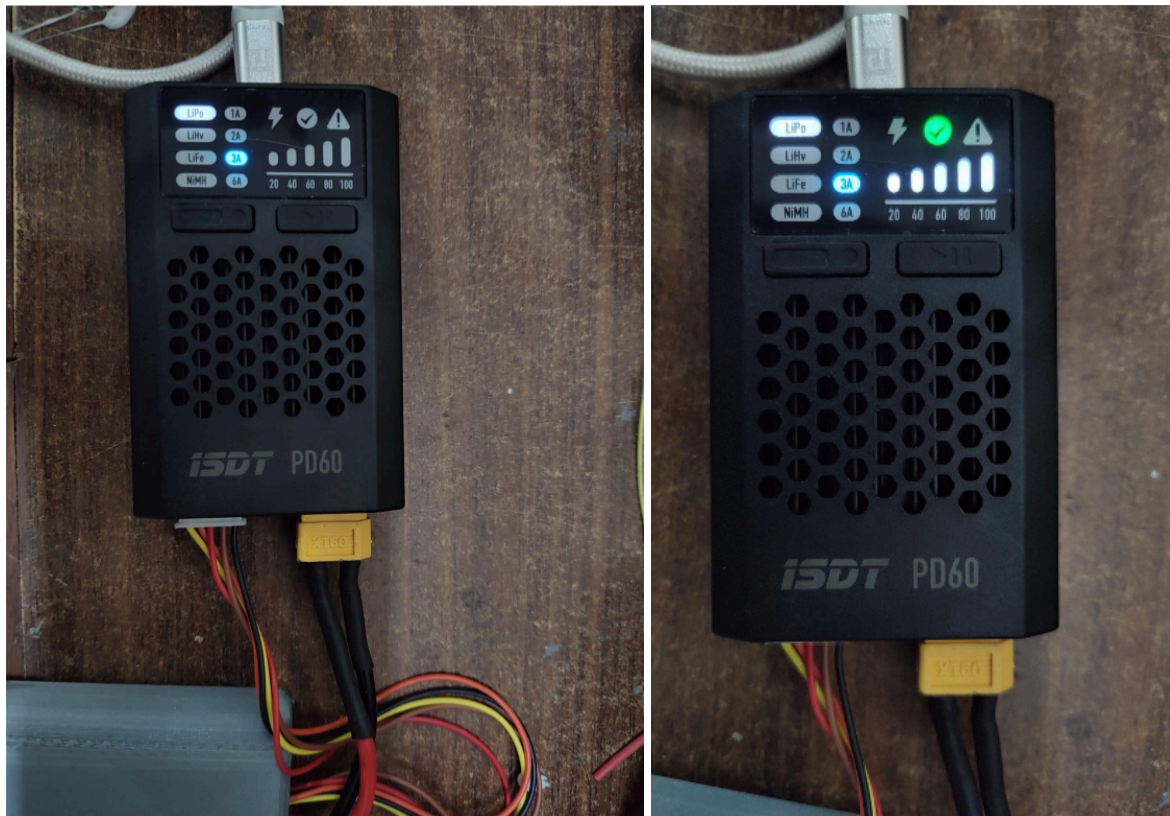
# Settings under path "all_others" are applied to all paths that
# do not match another entry.
all_others:
```

Overlaying telemetries on drone's feed

1. Run mediamtx in the command prompt and keep it open to keep the server running.
2. Make sure MAVLink forwarding is enabled. Run drone_hud.py to create hud_data.txt which will have all the important telemetries.
3. Run the ffmpeg command mentioned above.

Drone Battery Charging

1. Use the charger shown below.
2. The battery that comes with the drone allows 4A of current but since this charger does not have that option use 3A instead. Press the left button to change the current and long press the left button to change the battery chemistry. Use the LiPo option.
3. Press the right button to start charging.
4. Once the battery is fully charged the green symbol will light up.



Warning

Do not perform firmware updates via QGroundControl. The warning below was provided by the support team of Menthosa Solutions:

The Suparna drone runs a customized version of ArduPilot, and the update process is restricted. Attempting to upgrade will fail and may lead to the corruption of the Flight Controller. In the event that the firmware is compromised, restoration can only be performed by our team.

If you require low-level access to the onboard computer and the Flight Controller, we can provide SSH credentials that grant full access to the flight control stack. To initiate this process, please note the following:

1. NDA Requirement: You must sign a Non-Disclosure Agreement (NDA). This agreement specifies that any crashes following the sharing of credentials will not be covered under warranty, as we will be unable to determine if an issue resulted from a system fault or your specific code and configuration changes.
2. Official Request: You must send a request for these credentials using an official institutional email address, after which we will initiate the process by sending a NDA; personal email accounts will not be accepted.

In case of any queries or problems contact contactus@menthosa.com they were very helpful throughout this project.